
Vehicle Market Analytics System

Data Population — DML Statements & Validation

Milestone 5 — INSERT, UPDATE, DELETE and Validation

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Data Summary

497	12	PakWheels
Records	Tables	Dataset

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1. Introduction

This document covers Milestone 5: populating **pakwheels_db** with data from the cleaned Pak-Wheels dataset (497 records), executing UPDATE and DELETE operations, and running validation queries to verify row counts, NULL integrity, and foreign key consistency. All queries are standard MySQL, executable directly in MySQL Workbench.

2. INSERT Statements

Parent tables are populated before child tables to satisfy foreign key constraints.

2.1 locations

```
USE pakwheels_db;

INSERT INTO locations (city, state, country) VALUES
('Lahore',      'Punjab',      'Pakistan'),
('Karachi',     'Sindh',       'Pakistan'),
('Islamabad',   'ICT',         'Pakistan'),
('Rawalpindi',  'Punjab',      'Pakistan'),
('Faisalabad',  'Punjab',      'Pakistan'),
('Multan',      'Punjab',      'Pakistan'),
('Peshawar',    'Khyber Pakhtunkhwa', 'Pakistan'),
('Sargodha',    'Punjab',      'Pakistan'),
('Sialkot',     'Punjab',      'Pakistan'),
('Bahawalpur',  'Punjab',      'Pakistan');
```

2.2 categories

```
INSERT INTO categories (category_name, description) VALUES
('Sedan',      'Standard passenger sedans'),
('Hatchback',  'Compact hatchback vehicles'),
('SUV',        'Sport utility and crossover vehicles'),
('Van',        'Minivans and cargo vans'),
('Other',      'Pickup trucks, coupes, and convertibles');
```

2.3 users

```
INSERT INTO users (name, email, password_hash, phone, role) VALUES
('Admin User', 'admin@pakwheels.pk',
 '$2b$12$adminHash001', '03001000001', 'admin'),
('Ahmed Khan', 'ahmed@example.pk',
 '$2b$12$ahmedHash002', '03001000002', 'seller'),
('Sara Malik', 'sara@example.pk',
 '$2b$12$saraHash003', '03001000003', 'seller'),
('Bilal Raza', 'bilal@example.pk',
 '$2b$12$bilalHash004', '03001000004', 'buyer'),
('Hina Baig', 'hina@example.pk',
 '$2b$12$hinaHash005', '03001000005', 'buyer');
```

2.4 admins

```
INSERT INTO admins (user_id, permissions) VALUES
(1, '{"manage_listings": true, "manage_users": true}');
```

2.5 sellers

```
INSERT INTO sellers
(user_id, location_id, business_name, seller_type, verified_status)
VALUES
(2, 1, 'Ahmed Motors Lahore', 'dealer', 1),
(3, 3, 'Sara Cars Islamabad', 'individual', 0);
```

2.6 vehicles

The following records are taken directly from the cleaned PakWheels CSV dataset.

```
INSERT INTO vehicles
(brand, model, year, price, mileage, fuel_type,
transmission, body_type, engine_cc, color,
seller_id, location_id, category_id)
VALUES
('Toyota', 'Prado', 2016, 19500000, 97173,
'Petrol', 'Automatic', 'Suv', 2700, 'Unlisted', 1, 5, 3),
('Toyota', 'Prius', 2007, 3200000, 215099,
'Hybrid', 'Automatic', 'Sedan', 1500, 'Silver Metallic', 1, 5, 1),
('Toyota', 'Corolla', 2005, 2020000, 79284,
'Petrol', 'Manual', 'Sedan', 1800, 'Black', 1, 1, 1),
('Toyota', 'Passo', 2022, 4850000, 11400,
'Petrol', 'Automatic', 'Hatchback', 1000, 'Brown', 2, 3, 2),
('Honda', 'Civic', 2017, 4900000, 112000,
'Petrol', 'Automatic', 'Sedan', 1800, 'Taffeta White', 2, 3, 1),
('Honda', 'City', 2000, 1100000, 10000,
'Petrol', 'Manual', 'Sedan', 1300, 'Crystal Black', 1, 1, 1),
('Honda', 'Civic', 2021, 6300000, 70000,
'Petrol', 'Automatic', 'Sedan', 1800, 'Meteoroid Gray', 2, 6, 1),
('Toyota', 'Prius', 2014, 4950000, 88500,
'Hybrid', 'Automatic', 'Sedan', 1800, 'Black', 1, 2, 1),
('Changan', 'Karvaan', 2026, 3650000, 10000,
'Petrol', 'Manual', 'Van', 1200, 'White', 2, 5, 4),
('Daihatsu', 'Hijet', 2019, 2325000, 122000,
'Petrol', 'Automatic', 'Van', 660, 'Silver', 2, 2, 4),
('Kia', 'Sportage L', 2025, 11300000, 16000,
'Hybrid', 'Automatic', 'Suv', 1600, 'Clear White', 2, 3, 3),
('Honda', 'City', 2004, 1490000, 170000,
'Petrol', 'Manual', 'Sedan', 1300, 'Urban Titanium', 1, 2, 1),
('Honda', 'Civic', 2021, 6650000, 30000,
'Petrol', 'Automatic', 'Sedan', 1800, 'White', 1, 2, 1),
('Honda', 'Civic', 2015, 3950000, 105000,
'Petrol', 'Automatic', 'Sedan', 1800, 'Albatar Silver', 2, 9, 1),
('Suzuki', 'Alto', 2022, 1850000, 12000,
'Petrol', 'Manual', 'Hatchback', 660, 'Red', 2, 3, 2);
```

2.7 vehicle_images

```
INSERT INTO vehicle_images (vehicle_id, image_url, is_primary)
VALUES
(1, 'https://cdn.pakwheels.com/ads/prado-2016.jpg', 1),
(2, 'https://cdn.pakwheels.com/ads/prius-2007.jpg', 1),
(3, 'https://cdn.pakwheels.com/ads/corolla-2005.jpg', 1),
(4, 'https://cdn.pakwheels.com/ads/passo-2022.jpg', 1),
(5, 'https://cdn.pakwheels.com/ads/civic-2017.jpg', 1),
```

```
(6, 'https://cdn.pakwheels.com/ads/city-2000.jpg', 1),
(7, 'https://cdn.pakwheels.com/ads/civic-2021a.jpg', 1),
(8, 'https://cdn.pakwheels.com/ads/prius-2014.jpg', 1),
(9, 'https://cdn.pakwheels.com/ads/karvaan-2026.jpg', 1),
(10, 'https://cdn.pakwheels.com/ads/hijet-2019.jpg', 1);
```

2.8 favorites, inquiries, reviews, comparisons, search_logs

```
INSERT INTO favorites (user_id, vehicle_id) VALUES
(4, 1), (4, 5), (5, 3);

INSERT INTO inquiries (user_id, vehicle_id, seller_id, message) VALUES
(4, 1, 1, 'Is the Toyota Prado still available?'),
(5, 5, 2, 'Can I arrange a test drive for the Civic?'),
(4, 3, 1, 'What is the final price for the Corolla?');

INSERT INTO reviews (reviewer_id, seller_id, rating, comment) VALUES
(4, 1, 5, 'Very professional. Smooth transaction.'),
(5, 2, 4, 'Good experience. Car was as described.');
```

```
INSERT INTO comparisons (user_id, vehicle_id_1, vehicle_id_2) VALUES
(4, 1, 8),
(5, 5, 7);

INSERT INTO search_logs (user_id, query_text, results_count) VALUES
(4, 'Toyota Corolla', 18),
(5, 'Honda Civic', 24),
(4, 'Suzuki Alto', 11);
```

3. UPDATE Queries

```
-- Mark a vehicle as sold
UPDATE vehicles
SET status = 'sold'
WHERE vehicle_id = 9;

-- Update seller rating after a new review
UPDATE sellers
SET rating = (SELECT ROUND(AVG(rating), 2)
              FROM reviews
              WHERE seller_id = 1)
WHERE seller_id = 1;

-- Mark an inquiry as read
UPDATE inquiries
SET status = 'read'
WHERE inquiry_id = 2;

-- Set featured status for recent Toyota listings
UPDATE vehicles
SET status = 'featured'
WHERE brand = 'Toyota'
AND year >= 2020;
```

4. DELETE Queries

```
-- Remove a specific saved favorite
DELETE FROM favorites
WHERE user_id = 4 AND vehicle_id = 5;

-- Delete a specific inquiry
DELETE FROM inquiries
WHERE inquiry_id = 3;

-- Remove search logs older than 90 days
DELETE FROM search_logs
WHERE searched_at < NOW() - INTERVAL 90 DAY;
```

5. Validation Queries

5.1 Row Count per Table

```
SELECT COUNT(*) AS locations_count FROM locations;
SELECT COUNT(*) AS categories_count FROM categories;
SELECT COUNT(*) AS users_count FROM users;
SELECT COUNT(*) AS admins_count FROM admins;
SELECT COUNT(*) AS sellers_count FROM sellers;
SELECT COUNT(*) AS vehicles_count FROM vehicles;
SELECT COUNT(*) AS vehicle_images_count FROM vehicle_images;
SELECT COUNT(*) AS favorites_count FROM favorites;
SELECT COUNT(*) AS comparisons_count FROM comparisons;
SELECT COUNT(*) AS inquiries_count FROM inquiries;
SELECT COUNT(*) AS reviews_count FROM reviews;
SELECT COUNT(*) AS search_logs_count FROM search_logs;
```

Expected output after all DML:

Query	Expected Result
locations_count	10
categories_count	5
users_count	5
admins_count	1
sellers_count	2
vehicles_count	15
vehicle_images_count	10
favorites_count	2
comparisons_count	2
inquiries_count	2
reviews_count	2
search_logs_count	3

Table 1: Expected row counts after all INSERT, UPDATE, and DELETE operations

5.2 NULL Checks on Critical Columns

All queries below must return 0.

```

SELECT COUNT(*) AS null_brand      FROM vehicles WHERE brand      IS
NULL;
SELECT COUNT(*) AS null_model      FROM vehicles WHERE model      IS
NULL;
SELECT COUNT(*) AS null_price      FROM vehicles WHERE price      IS
NULL;
SELECT COUNT(*) AS null_seller     FROM vehicles WHERE seller_id   IS
NULL;
SELECT COUNT(*) AS null_email      FROM users    WHERE email      IS
NULL;
SELECT COUNT(*) AS null_password   FROM users    WHERE password_hash IS
NULL;
SELECT COUNT(*) AS null_message    FROM inquiries WHERE message    IS
NULL;
SELECT COUNT(*) AS invalid_rating  FROM reviews
WHERE rating NOT BETWEEN 1 AND 5;

```

5.3 Foreign Key Integrity Checks

All queries below must return 0.

```

-- Vehicles with no matching seller
SELECT COUNT(*) AS orphan_vehicles
FROM   vehicles v
LEFT   JOIN sellers s ON v.seller_id = s.seller_id
WHERE  s.seller_id IS NULL;

-- Sellers with no matching user
SELECT COUNT(*) AS orphan_sellers
FROM   sellers s
LEFT   JOIN users u ON s.user_id = u.user_id
WHERE  u.user_id IS NULL;

-- Images with no matching vehicle
SELECT COUNT(*) AS orphan_images
FROM   vehicle_images vi
LEFT   JOIN vehicles v ON vi.vehicle_id = v.vehicle_id
WHERE  v.vehicle_id IS NULL;

-- Inquiries with no matching vehicle or seller
SELECT COUNT(*) AS orphan_inquiries
FROM   inquiries i
LEFT   JOIN vehicles v ON i.vehicle_id = v.vehicle_id
LEFT   JOIN sellers  s ON i.seller_id  = s.seller_id
WHERE  v.vehicle_id IS NULL OR s.seller_id IS NULL;

```

5.4 SELECT Queries for Verification

```

-- All vehicle listings with seller name and city
SELECT v.brand, v.model, v.year, v.price,
       u.name AS seller,
       l.city AS city
FROM   vehicles v

```

```
JOIN    sellers    s ON v.seller_id    = s.seller_id
JOIN    users      u ON s.user_id      = u.user_id
JOIN    locations  l ON v.location_id  = l.location_id
ORDER   BY v.price DESC;

-- Average price per brand
SELECT  brand,
        COUNT(*)           AS total_listings,
        ROUND(AVG(price), 0) AS avg_price
FROM    vehicles
WHERE   status != 'sold'
GROUP   BY brand
ORDER   BY avg_price DESC;

-- Active listings count per seller
SELECT  u.name AS seller_name,
        s.rating,
        COUNT(v.vehicle_id) AS active_listings
FROM    sellers s
JOIN    users    u ON s.user_id    = u.user_id
LEFT    JOIN    vehicles v ON v.seller_id = s.seller_id
        AND v.status != 'sold'
GROUP   BY s.seller_id;
```